

2021 秋 A 卷参考答案

一、 1.-3 2. 3 3. $\begin{pmatrix} -9 & -1 & 1 \\ 12 & 1 & 3 \\ 11 & 1 & -1 \end{pmatrix}$ 4. -16 5. $k(-1, 1, -1, -2)^T$ 6. 15

二、 1-6 BACBAA

三、 1.
$$\begin{vmatrix} -1 & -1 & \cdots & -1 & n \\ -1 & -1 & \cdots & n & -1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & n & \cdots & -1 & -1 \\ n & -1 & \cdots & -1 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & \cdots & 1 & 1 \\ -1 & -1 & \cdots & n & -1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & n & \cdots & -1 & -1 \\ -1 & -1 & \cdots & -1 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & \cdots & 1 & 1 \\ 0 & 0 & \cdots & n+1 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & n+1 & \cdots & 0 & 0 \\ n+1 & 0 & \cdots & 0 & 0 \end{vmatrix} = (-1)^{\frac{n(n-1)}{2}} (n+1)^{n-1}$$

2.
$$\begin{pmatrix} 1 & 0 & 1 & 3 & 2 \\ -1 & 3 & -1 & 0 & 1 \\ 2 & 1 & 2 & 7 & 5 \\ 0 & 2 & 4 & 14 & 6 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & 3 & 2 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

向量组的秩为 3. $\alpha_1, \alpha_2, \alpha_3$ 为向量组的一个极大无关组, $\alpha_4 = \alpha_2 + 3\alpha_3$, $\alpha_5 = \alpha_1 + \alpha_2 + \alpha_3$.

3. (1) 设所求矩阵为 A, 则 $\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} A = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$

$\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 3 & -1 & 0 \\ 0 & -1 & 3 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 8 & 3 \\ 0 & 1 & -3 & -1 \end{pmatrix}$, 所以 $A = \begin{pmatrix} 8 & 3 \\ -3 & -1 \end{pmatrix}$.

(2) $\alpha = -\begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

设所求坐标为 $(x_1, x_2)^T$, 则 $x_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} + x_2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$, 得 $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$.

4. $A\alpha = \lambda\alpha$ 得 $a = -3, b = 0, \lambda = -1$.

四、证明: 设 $k_1\beta_1 + k_2\beta_2 + k_3\beta_3 = 0$,

即 $k_1(2\alpha_1 + \alpha_2) + k_2(3\alpha_2 + \alpha_3) + k_3(\alpha_1 + 4\alpha_3) = 0$

整理得

$$(2k_1 + k_3)\alpha_1 + (k_1 + 3k_2)\alpha_2 + (k_2 + 4k_3)\alpha_3 = 0,$$

因为向量组 $\alpha_1, \alpha_2, \alpha_3$ 线性无关,

所以 $\begin{cases} 2k_1 + k_3 = 0 \\ k_1 + 3k_2 = 0 \\ k_2 + 4k_3 = 0 \end{cases}$ 因 $\begin{vmatrix} 2 & 0 & 1 \\ 1 & 3 & 0 \\ 0 & 1 & 4 \end{vmatrix} = 25 \neq 0$, 所以 $k_1 = k_2 = k_3 = 0$, 从而向量组 $\beta_1, \beta_2, \beta_3$

线性无关.

$$\text{五、} \begin{pmatrix} 1 & 0 & 4 & -1 & -1 \\ 1 & 1 & -4 & 1 & 6 \\ 2 & 1 & a-1 & b-3 & b+6 \\ -2 & -1 & 0 & b-2 & b-2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 4 & -1 & -1 \\ 0 & 1 & -8 & 2 & 7 \\ 0 & 0 & a-1 & b-3 & b+1 \\ 0 & 0 & 0 & b-2 & b+3 \end{pmatrix}$$

当 $a \neq 1, b = 2$ 时, 方程组无解; 当 $a \neq 1, b \neq 2$ 时, 方程组有唯一解.

$$\text{当 } a = 1 \text{ 时, } \begin{pmatrix} 1 & 1 & -4 & 1 & 6 \\ 0 & 1 & -8 & 2 & 7 \\ 0 & 0 & 0 & b-3 & b+1 \\ 0 & 0 & 0 & b-2 & b+3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & -4 & 1 & 6 \\ 0 & 1 & -8 & 2 & 7 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 7-b \end{pmatrix}$$

当 $a = 1, b \neq 7$ 时, 方程组无解;

当 $a = 1, b = 7$ 时, 方程组有无穷多解.

综上, 当 $b = 2$ 或 $\begin{cases} a=1 \\ b \neq 7 \end{cases}$ 时, 方程组无解; 当 $\begin{cases} a \neq 1 \\ b \neq 2 \end{cases}$ 时, 方程组有唯一解; 当 $\begin{cases} a=1 \\ b=7 \end{cases}$ 时, 方程组有无穷多解.

$$\begin{pmatrix} 1 & 1 & -4 & 1 & 6 \\ 0 & 1 & -8 & 2 & 7 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 4 & 0 & 1 \\ 0 & 1 & -8 & 0 & 3 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

可取自由变量 x_3 ,

$$(1) Ax = b \text{ 特解为: } \xi_0 = \begin{pmatrix} 1 \\ 3 \\ 0 \\ 2 \end{pmatrix}, (2) Ax = 0 \text{ 基础解系为: } \xi_1 = \begin{pmatrix} -4 \\ 8 \\ 1 \\ 0 \end{pmatrix},$$

$$(3) Ax = b \text{ 一般解 } \xi = \begin{pmatrix} 1 \\ 3 \\ 0 \\ 2 \end{pmatrix} + k \begin{pmatrix} -4 \\ 8 \\ 1 \\ 0 \end{pmatrix}, \text{ 其中 } k \text{ 为任意常数.}$$

$$\text{六、} A = \begin{pmatrix} 2 & 1 & -4 \\ 1 & -1 & 1 \\ -4 & 1 & a \end{pmatrix}$$

1. 因为有一个特征值是 0, 所以 $|A| = 0 \Rightarrow a = 2$.

$$2. |\lambda I - A| = \begin{vmatrix} \lambda-2 & -1 & 4 \\ -1 & \lambda+1 & -1 \\ 4 & -1 & \lambda-2 \end{vmatrix} = (\lambda-6)(\lambda+3)\lambda, \text{ 故特征值是 } 6, -3, 0$$

$$\text{当 } \lambda = 6 \text{ 时, } 6I - A = \begin{pmatrix} 4 & -1 & 4 \\ -1 & 7 & -1 \\ 4 & -1 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \text{ 得 } a_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \text{ 单位化得 } \eta_1 = \frac{1}{\sqrt{2}}(-1, 0, 1)^T,$$

$$\text{当 } \lambda = -3 \text{ 时, } -3I - A = \begin{pmatrix} -5 & -1 & 4 \\ -1 & -2 & -1 \\ 4 & -1 & -5 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \text{ 得 } a_2 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \text{ 单位化得 } \eta_2 = \frac{1}{\sqrt{3}}(1, -1, 1)^T,$$

$$\text{当 } \lambda = 0 \text{ 时, } 0I - A = \begin{pmatrix} -2 & -1 & 4 \\ -1 & 1 & -1 \\ 4 & -1 & -2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}, \text{ 得 } a_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \text{ 单位化得 } \eta_3 = \frac{1}{\sqrt{6}}(1, 2, 1)^T,$$

故得正交矩阵 $Q = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{pmatrix}$.

3. 规范形为 $z_1^2 - z_2^2$.